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PROJECT 2

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Abstract

This report will conduct a trend and ratio analysis on Pick N Pay group for the 2020-year comparative to the 2019-year. The objective of this analysis is to establish the health of the company and areas for potential improvement. The trends company's income statement and balance sheets trends will be found through the utilization of horizontal and vertical analysis. Furthermore, the health of the company will be determined through ratio analysis. The results of the analysis are for the use of internal and external stakeholders in order to make decisions for the company.

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Introduction

Financial statements give internal and external stakeholders an indication of the overall performance of a business (OpenStax, 2023). Informed business decisions are conducted from the results of the analysis of financial trends and ratios that can be extracted from the abovementioned financial statements. Additionally, areas for improvement can be exposed. The objective of this report is to analyze financial information extracted from audited financial statements for a year-on-year comparison of Pick N Pay Group for the years 2020 and 2019. The financial health of the company will be investigated through a horizontal and vertical trend analysis as well as ratio analysis (OpenStax, 2023).

Horizontal analysis

The financial statements of Pick N Pay (Olivier, 2023) for 2020 and 2019 were utilized for trend and ratio analysis. The horizontal comparisons were calculated from the main headings, as well as relevant line items, in the Statement of Financial Position and Income Statement.
The percentage change between the two years was calculated using the following equations:

- $Rand\ Change = Year\ of\ Analysis\ Amount - Base\ Year\ Amount$
- $Percentage\ Change = \frac{Rand\ Change}{Base\ Year\ Amount} \times 100$

The comparative year-end income statement horizontal analysis yielded results as follows (Olivier, 2020). It was found that net sales had 1.12% increase. The PNP group observed a gross profit increase of 0.34%. Net sales and gross profit illustrated a positive trend. For the 2020 year, the group had a net income decrease of 17.30%, as well as a 17.48% decrease in earnings per share. Therefore, this indicates a negative trend.
The comparative year-end balance sheets horizontal analysis yielded the following results (Olivier, L. 2020). Accounts receivable decreased by 3.09%. This negative trend could have been due to the coronavirus pandemic that caused consumers to lose their jobs and therefore unable to pay. Although, the PNP group may remedy this by introduce stricter lending policies. Accounts payable increased by 8.78%. Total assets and total liabilities and stockholder equity both saw a 4.35% increase. The abovementioned line items saw a positive trend due to the negative effects of the coronavirus pandemic.

Vertical Analysis

The vertical analysis was conducted using values that were extracted from the year-end Statement of Financial Position and The Statement of Comprehensive Income, and can be seen in Appendix B. The following equation was used to calculate common size percentage:

- $Common\ Size\ Percentaage = \frac{Comparison\ amount}{Base\ Amount} \times 100$

The results obtained from the comparative year-end income statement's vertical analysis were found to be as follows (Olivier, L.). Gross profit was 20% of the net sales balance which saw an increase of 1% from the previous year. The cost of goods sold was 80% of the net

sales balance which was a decrease from the previous year. Trading expense and trading profit each increased by 1% from the previous year with 18% and 4% of the net sales balance, respectively. Therefore, a positive trend was observed in gross profit and trading profit. Moreover, the cost of goods sold had a negative trend which may be beneficial for the company's and investors' gains.
The comparative year-end balance sheet's vertical analysis was observed as follows(Olivier, L.2020). Accounts receivable decreased by 1% from the previous year and made up 12% of total assets. This positive trend may be remedied by stricter lending policies in future. Inventory and total current assets increased by 1% from the previous year and made up 19% and 38% of the total assets, respectively. This positive trend may indicate growth in the business. Accounts payable saw a 2% increase and made up 34% of total liabilities and stockholder equity. Total liabilities and total stockholder equity made up 91% and 9% of total liabilities and stockholder equity, respectively. Total liabilities had a 0.5% increase whereas total stockholder equity had a 0.5% decrease. The positive trend of accounts payable, stockholder equity and total liabilities may be due to the coronavirus pandemic that occurred in 2020.

Ratio analysis

Liquidity ratios

The liquidity for the 2020 year of the PNP group was investigated. It was found that the company did not have a strong ability to pay off its short-term obligations; this indicates bad financial health. This was confirmed by the working capital, current ratio, and quick ratio calculations below.

- $Working\ Capital = Current\ Assets - Current\ Liabilities$

$$= 12886.7 - 16303.65 \quad = \mathbf{-3416.95}$$

The Pick N Pay's current assets are less than its current liabilities, which shows that it would not be able to cover short-term debt.

- $Current\ Ratio = \frac{Current\ Assets}{Current\ Liabilities} = \frac{12886.7}{16303.65} = \mathbf{0.7904}$

A 0.7904:1 ratio depicts the ability of the company to cover obligations; in this instance, Pick N Pay is unable because it has 0.79 as many current assets as current liabilities.

$$Quick\ Ratio = \frac{Cash + Short - Term\ Investments + Accounts\ Receivable}{Current\ Liabilities}$$

$$= \frac{1947.3+221+4168.5}{16303.65} = \mathbf{0.3887}$$

A 0.39:1 ratio illustrates that the company does not have an adequate amount of quick assets to cover immediate obligations.

Solvency ratios

These ratios illustrated that the PNP group is able to meet its long-term obligations. Although the debt-to-equity ratio is not ideal, it may be skewed because of the effect of the pandemic that occurred in 2020.

- $Debt\ To\ Equity\ Ratio = \frac{Total\ Liabilities}{Total\ Stockholder\ Equity} = \frac{30495.2}{3010.1} = \mathbf{10.13}$

For every R1 of equity contributed to financing, R10.13 is contributed to lenders. Potential investors could interpret this as movement to insolvency. Getting the ratio to 1:1 would improve long-term business viability.

- $Times\ Interest\ Earned\ Ratio$
$$= \frac{Earnings\ before\ Interest\ and\ Taxes}{Interest\ Expense}$$
$$= \frac{1913.9}{1772.4} = \mathbf{1.08\ times}$$

This 1.08 times indicates the ease with which Pick N Pay can repay interest on long-term outstanding loans.

Efficiency ratios

The PNP group managed and utilized its assets well in 2020. This is confirmed by the ratios below.

- $Average\ Accounts\ Receivable$
$$= \frac{Beginning\ Accounts\ Receivable + Ending\ Accounts\ Receivable}{2}$$
$$= \frac{4301.4+4168.5}{2} = \mathbf{4234.95}$$
- $Accounts\ Receivable\ Turnover = \frac{Net\ Credit\ Sales}{Average\ Accounts\ Receivable}$
$$= \frac{4168.5}{4234.95} = \mathbf{0.9843\ times}$$

Accounts receivable turnover of 0.98 times per year exhibits a low rate of collection on outstanding amounts. Stricter lending policies will increase the accounts receivable.

- $Average\ Total\ Assets = \frac{Beginning\ Total\ Assets + Ending\ Total\ Assets}{2}$
$$= \frac{33505.3+32107.7}{2} = 32806.5$$
- $Total\ Asset\ Turnover = \frac{Net\ Sales}{Average\ Total\ Assets} = \frac{89281.5}{32806.5} = \mathbf{2.72\ times}$

For every R1 of assets, R2.72 of net sales are generated. It is good to see this value increase over time.

- $Average\ Inventory = \frac{Beginning\ Inventory + Ending\ Inventory}{2}$
$$= \frac{6519.8 + 5693.4}{2}$$
$$= 6106.6$$
- $Inventory\ Turnover = \frac{Cost\ of\ Goods\ Sold}{Average\ Inventory}$
$$= \frac{71679.8}{6106.6}$$

$$=\mathbf{11.74\ times}$$

Pick N Pay changes its inventory 11.74 times a year.

- $Days'Sales\ in\ Inventory = \frac{Ending\ Inventory}{Cost\ of\ Goods\ Sold} \times 365$
  $= \frac{6519.8}{71679.8} \times 365$
  $= \mathbf{33\ days}$

It takes the company 33 days to sell inventory.

Profitability ratios

The company exhibits a low profit margin and return on total assets, which may be due to the 2020 coronavirus pandemic (Olivier, 2020). However, it had a good return on investments as is shown in the following calculations.

- $Profit\ Margin = \frac{Net\ Income}{Net\ Sales} = \frac{1194.7}{89281.5}$ $=\mathbf{0.013}$

For every R1 of sales, there is R0.013 return on profit. Pick N Pay would need to increase its sales and decrease its expenses to improve the return on profit (OpenStax, 2023). A low rate of sales may be a result of lockdown during the 2020 pandemic.

- $Return\ on\ Total\ Assets = \frac{Net\ Income}{Average\ Total\ Assets}$
  $= \frac{1194.7}{32806.5} = \mathbf{0.036}$

Return on total assets is 3.6%; this value is too low to be acceptable. Therefore, the company is not doing well at utilizing its assets to generate a profit.

- $Average\ Stockholder\ Equity$
  $= \frac{Beginning\ Stockholder\ Equity + Ending\ Stockholder\ Equity}{2}$
  $= \frac{3035+3010.1}{2}$
  $=3022.55$

- $Return\ on\ Equity = \frac{Net\ Income}{Average\ Stockholder\ Equity}$
  $= \frac{1194.7}{3022.55}$
  $= \mathbf{0.395}$

Pick N Pay has a return on investment of 39.5%. Therefore, the company makes a good profit from investments.

Conclusion

Financial statement analysis is essential for making decisions for a company. This analysis produces information about the overall performance of the company and where necessary improvements can be made. The PNP group's financial statements were analyzed, and it was found that the company had a current ratio of 0.79 and an even lower quick ratio of 0.39. This was strong evidence showing that the company could not meet short-term obligations. However, the solvency ratio depicted that the PNP group would be able to meet its long-

term obligations 1.08 times interest. The efficiency ratios supported that the company managed its assets very well in 2020, with 2.72 times asset turnover and 11.4 times inventory turnover. The profitability ratios supported that the PNP group had a good return on investments of 39.5%. The shortcomings of the company can be deduced has having been a result of lockdown during the coronavirus pandemic. Furthermore, the company shows promise to grow and improve in future. Investments should be made in future in order to support the growth of this company.

References

Olivier, L. 2020*. Pick N Pay Audited Annual Financial Statements 2020*. South Africa.

OpenStax. 2023. *Principles of Accounting.* Available: http://cnx.org/content/col25448/1.4 [17 May 2023].

Appendices

Appendix A: Financial statements and horizontal analysis

Comparative year-end income statements horizontal analysis				*Comparative year-end balance sheets horizontal analysis*			
	2020	**2019**	**% change**		**2020**	**2019**	**% change**
				Assets:			
Net sales	89281.5	88293.2	1.12	Cash	1947.3	1503.2	29.54
Cost of Merchandise Sold	71679.8	71436.7	0.34	Accounts receivable	4168.5	4301.4	-3.09
Gross Profit	17601.7	16856.5	4.42	Inventory	6519.8	5693.4	14.52
Trading Expense	16023.9	15299.5	4.73	Short-term investments	221	248.9	-11.21
Trading profit	3148	3054.9	3.05	Total current assets	12886.7	11770.6	9.48
Interest expense	1772.1	1726.4	2.65	Total assets	33505.3	32107.7	4.35
Profit before tax before capital items	1913.9	1883	1.64	**Liabilities:**			
Tax	541.3	463.8	16.71	Accounts Payable	11255.2	10346.3	8.78
Profit before tax	1736	1908.4	-9.03	Unearned revenue	298.8	256.2	16.63
Net income	1194.7	1444.6	-17.30	Total current liabilities	16303.65	15423.4	5.71
Total comprehensive income for period	1157	1166.7	-0.83	Total liabilities	30495.2	29072.7	4.89
Earnings per share	250.9	304.04	-17.48	**Stockholder Equity:**			
				Ending retained earnings	4303.2	4331.9	-0.66
				Total stockholder equity	3010.1	3035	-0.82
				Total Liabilities and stockholder equity	33505.3	32107.7	4.35

Appendix B: Financial statements and vertical analysis

Comparative year-end income statements vertical analysis

	2020	**2020**	**2019**	**2019**
Net sales	89281.5	100%	88293.2	100%
Cost of Goods Sold	71679.8	80%	71436.7	81%
Gross Profit	17601.7	20%	16856.5	19%
Trading Expense	16023.9	18%	15299.5	17%
Trading profit	3148	4%	3054.9	3%
Interest expense	1772.1	2%	1726.4	2%
Profit before tax before capital items	1913.9	2%	1883	2%
Tax	541.3	1%	463.8	0.5%
Profit before tax	1736	2%	1908.4	2%
Net income	1194.7	1%	1444.6	2%
Total comprehensive income for period	1157	1%	1166.7	1%
Earnings per share	250.9	0.3%	304.04	0.3%

Comparative year-end balance sheets vertical analysis

	2020	**2020 Common size**	**2019**	**2019 Common size**
Assets:				
Cash	1947.3	6%	1503.2	5%
Accounts receivable	4168.5	12%	4301.4	13%
Inventory	6519.8	19%	5693.4	18%
Short-term investments	221	1%	248.9	1%
Total current assets	12886.7	38%	11770.6	37%
Total assets	33505.3	100%	32107.7	100%
Liabilities:				
Accounts Payable	11255.2	34%	10346.3	32%
Unearned revenue	298.8	1%	256.2	1%
Total current liabilities	16303.65	49%	15423.4	48%
Total liabilities	30495.2	91%	29072.7	90.5%
Stockholder Equity:				
Ending retained earnings	4303.2	13%	4331.9	13.5%
Total stockholder equity	3010.1	9%	3035	9.5%
Total Liabilities and stockholder equity	33505.3	100%	32107.7	100%